Fiscal Year:	FY 2018	**Task Last Updated:**	FY 07/15/2018
PI Name:	Bryan, Noelle Ph.D.		
Project Title:	Genomic and Functional Analysis of Desiccation Tolerance of International Space Station Isolated Enterococcus faecalis and Their Pathogenicity in Caenorhabditis elegans Genomic and Functional Analysis of Biofilm Morphotypes of International Space Station Isolated Staphylococcus epidermidis and Their Pathogenicity in Caenorhabditis elegans)		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Microbiology		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Organization Name:	Massachusetts Institute of Technology		
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Zip Code:	02139-4301	**Congressional District:**	7
Comments:	Dr. Bryan moved to NASA Ames Research Center in December 2019.		
Project Type:	Ground	**Solicitation / Funding Source:**	2016 Space Biology (ROSBio) NNH16ZTT001N-MOBE. Appendix B: Post-Doctoral Fellowships to Study the Microbiome of the ISS as a Built Environment
Start Date:	09/13/2017	**End Date:**	09/12/2019
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA ARC
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Zuber, Maria Ph.D. (MENTOR/ Massachusetts Institute of Technology) Carr, Christopher Ph.D. (Massachusetts Institute of Technology)		
Grant/Contract No.:	80NSSC17K0688		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	POSTDOCTORAL FELLOWSHIP NOTE: This project is funded by NASA as a result of a Space Act Agreement between NASA and the Sloan Foundation, for Collaborative Research Proposal for Microbiology of the Built Environment (MoBE). The purpose of this Space Act Agreement is to enhance the ability of NASA and the Sloan Foundation to support scientific research and the dissemination of information about the effect of spaceflight on microbial ecosystems in spacecraft such as the International Space Station (ISS) through sponsorship of postdoctoral fellowships. NASA is providing the isolates for all studies. NEW TITLE--July 2018: Genomic and Functional Analysis of Desiccation Tolerance of International Space Station Isolated Enterococcus faecalis and Their Pathogenicity in Caenorhabditis elegans. Change in organism used in this project, approved by NASA Headquarters. Enterococcus faecalis is an opportunistic human pathogen consistently isolated and detected by molecular analysis on board the International Space Station (ISS). The ability of this organism to survive outside of the human gut may have important implications for its spread to a new host environment. The goal of this study is to characterize the desiccation survival capacity of ISS isolated E. faecalis strains and to investigate their pathogenicity in the model organism Caenorhabditis elegans. First, we shall provide the first comparative genomic information of E. faecalis strains to characterize strain-specific sequence variants and the identification of traits that may confer increased resistance to desiccation and antibiotics. We will also examine the presence of the clustered regularly interspaced short palindromic repeats-CRISPR-associated proteins (CRISPR-Cas) that confer bacterial resistance to foreign DNA and may provide an additional method of persistence to the transfer of mobile genetic elements. Phenotypic data will determine how desiccation tolerance varies, and how these variations relate to the clinical and commensal strain types. In vivo pathogenicity tests will culminate our investigation into the effect of E. faecalis biofilm morphotypes on the ability to establish infection in the model organism Caenorhabditis elegans. This work will help to determine genomic, phenotypic, and physiological basis for various desiccation and antibiotic resistance of ISS isolated E. faecalis, and provide insight to pathogenicity of spaceflight isolated microorganisms, where astronaut immunity is decreased and the availability of antibiotics is limited.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This project will investigate the potential for opportunistic human pathogens, E. faecalis, to survive outside of the human gut. These isolates have not been genetically or phenotypically characterized and those are the first steps of this project.
Task Progress:	While waiting on the arrival of the Enterococcus faecalis isolates recovered from the International Space Station (ISS), I began my fellowship research by expanding my skill set as a microbiologist. I was given the opportunity to analyze my first metagenomic data set of microbial communities collected from the soils of Vulcan Copahue, Chile. This introduction into bioinformatics would later prove to be a highly effective exercise for developing the skills needed to analyze the genomic data from the ISS strains. In addition, I spent this time learning the techniques necessary for library preparation and sequencing using the Nanopore MinION sequencer, the same technology recently used onboard the ISS. These skills, combined with my previous NASA flight operations experience, provided me with the opportunity to perform the first MinION DNA sequencing under simulated Mars and Europa gravity conditions while onboard a Zero-G parabolic flight (https://). The data from this flight have recently been accepted by npj Microgravity (Carr et al., 2018), and a second manuscript is currently in progress. Upon receiving the ISS strains the first objective was to characterize their genomic content, and to compare them to previously identified isolates. The assembly of each isolate's genome was accomplished by combining Illumina (250 nt, paired end reads) with Nanopore sequencing technologies. Using the tools provided by the Center for Genomic Epidemiology (CGE) website (http://www.genomicepidemiology.org), I was able to identify the sequence type, acquired antibiotic resistance genes, virulence genes, phage elements, plasmids, and the potential for these strains to serve as human pathogens. With the known sequence type, I was able to find similar strains with fully assembled genomes and reconstruct a phylogenetic tree to determine the closest relatives to the ISS strains. Additionally, by performing a pan-genome analysis, I was able to identify which genes are unique to the ISS strains. Briefly, ISS_1 (strain type (ST) 65) is most closely related to human commensal strains of E. faecalis, yet it also contains many previously documented virulence genes. Isolates ISS_2 and ISS_3 (ST30) were recovered on the same day at two different locations within the U.S. Lab. Analysis of the 2191 core genes shared by all E. faecalis isolates in this study revealed these isolates differ by only 2 single nucleotide polymorphisms. ISS_4 (ST40) is most closely related to isolate JH-1, and is a member of the clonal complex 40, a group of E. faecalis strains recovered from clinical settings. ISS_4 was also positive for all of the listed virulence traits for E. faecalis. All four strains tested positive (>80% likelihood) for the potential to serve as human pathogens, as predicted by the PathogenFinder feature of CGE. The genomic data is currently guiding the development of the second objective of my fellowship, the phenotypic characterization of the isolates and their potential for pathogenicity. Following the efforts of Schiwon et al., 2013, I will be determining the minimum inhibitory concentrations of different classes of antibiotics for each isolate. I will also characterize each isolate's ability to form a biofilm, a trait that may prove beneficial in a zero-gravity environment. Because each of the isolates were recovered from surface or air samples, I will investigate their ability to survive extended periods of desiccation. The ability of Enterococcus species to survive outside of the human gut may be a driving force behind hospital outbreaks, and equally important onboard the ISS. Each of these assays will be performed using E. faecalis OG1RF as a representative commensal-like strain, and E. faecalis MMH594 as a representative clinical isolate to provide a range of comparison for the ISS strains. Carr CE, Bryan NC, Saboda K, Bhattaru SA, Ruvkun G, Zuber MT. Acceleration Profiles and Processing Methods for Parabolic Flight. npj Microgravity (accepted 7/3/18). Preprint: arXiv:1712.05737 Schiwon K, Arends K, Rogowski KM, Fürch S, Prescha K, Sakinc T, Van Houdt R, Werner G, Grohmann E. Comparison of antibiotic resistance, biofilm formation and conjugative transfer of Staphylococcus and Enterococcus isolates from International Space Station and Antarctic Research Station Concordia. MicrobEcol. 2013 Apr;65(3):638-51.

Bibliography Type:	Description: (Last Updated: 02/01/2021)
Articles in Peer-reviewed Journals	Carr CE, Bryan NC, Saboda K, Bhattaru SA, Ruvkun G, Zuber MT. "Acceleration profiles and processing methods for parabolic flight." npj Microgravity. 2018 Aug 7;4:14. eCollection 2018. https://doi.org/10.1038/s41526-018-0050-3 ; PubMed PMID: 30109261; PubMed Central PMCID: PMC6081456 , Aug-2018